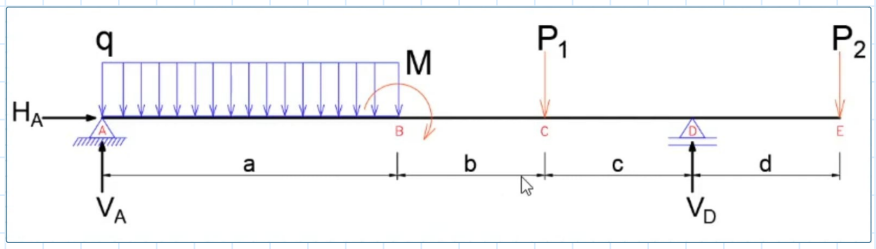
EJERCICIO 01

DATOS:

$q := 15\ \frac{kg}{m}$ $P_1 := 20\ kg$ $P_2 := 8\ kg$ $M := 10\ kg \cdot m$

$a := 2\ m$ $b := 1\ m$ $c := 1\ m$ $d := 1\ m$



CALCULO DE REACCIONES

Solución:

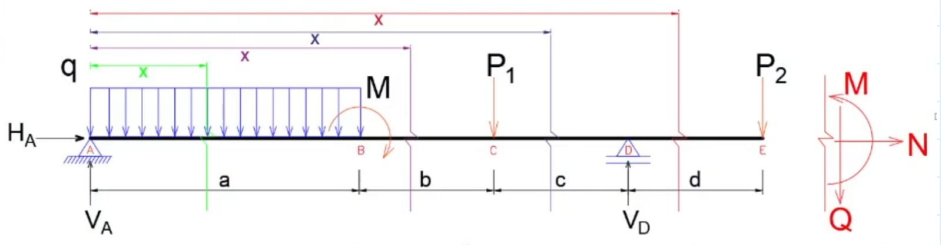
$\sum M_A = 0$ $V_D := q \cdot \frac{a^2}{2} + M + P_1 \cdot (a+b) - V_D \cdot (a+b+c) + P_2 \cdot (a+b+c+d) = 0 \xrightarrow{solve, V_D} 35 \cdot kg$

$V_D := 35\ kg$ $V_D = 35\ kg$

$\sum F_y = 0$ $V_A := V_A - q \cdot a - P_1 + V_D - P_2 = 0 \xrightarrow{solve, V_A} 23 \cdot kg$

$V_A := 23\ kg$ $V_A = 23\ kg$

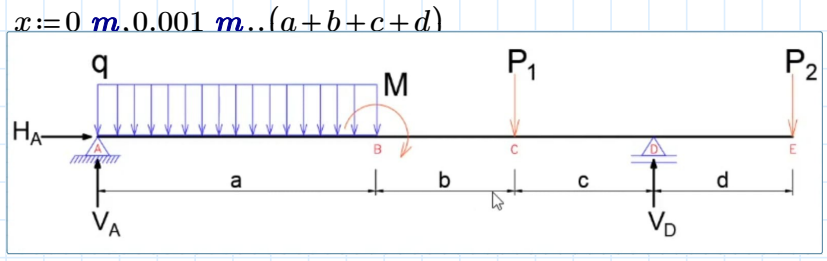
SECCIONES



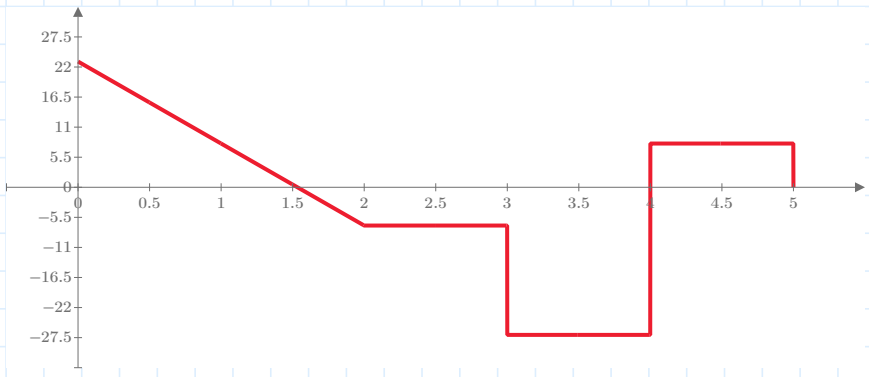
$M(x) := V_A \cdot x - (a > x > 0) \cdot q \cdot \frac{x^2}{2} - (x \geq a) \cdot q \cdot a \cdot \left(x - \frac{a}{2}\right) + (x \geq a) \cdot M - (x \geq (a+b)) \cdot P_1 \cdot (x - (a+b)) + (x \geq (a+b+c)) \cdot V_D \cdot (x - (a+b+c))$

$Q(x) := V_A - (a > x > 0) \cdot q \cdot x - (x \geq a) \cdot q \cdot a - (x \geq (a+b)) \cdot P_1 + (x \geq (a+b+c)) \cdot V_D - (x \geq (a+b+c+d)) \cdot P_2$

$x := 0\ m, 0.001\ m .. (a+b+c+d)$

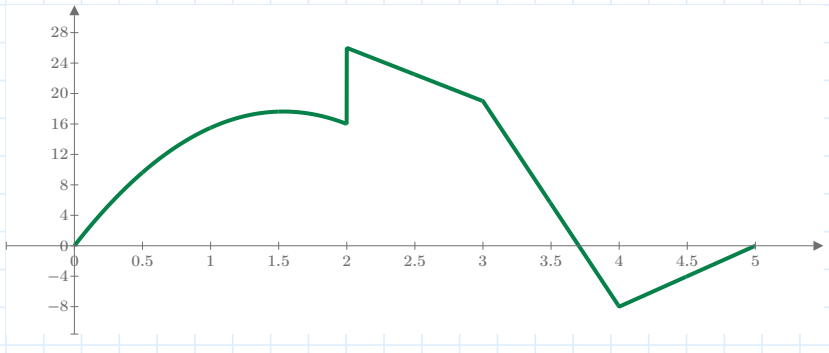


$M(x) = \begin{bmatrix} 0 \\ 0.023 \\ 0.046 \\ 0.069 \\ 0.092 \\ 0.115 \\ 0.138 \\ 0.161 \\ 0.184 \\ 0.206 \\ 0.229 \\ 0.252 \\ \vdots \end{bmatrix} kg \cdot m$ $Q(x) = \begin{bmatrix} 23 \\ 22.985 \\ 22.97 \\ 22.955 \\ 22.94 \\ 22.925 \\ 22.91 \\ 22.895 \\ 22.88 \\ 22.865 \\ 22.85 \\ 22.835 \\ \vdots \end{bmatrix} kg$



$Q(x)\ (kg)$

$x\ (m)$



$M(x)\ (kg \cdot m)$

$x\ (m)$